

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and deployed **production ML/GenAI systems** that delivered measurable business impact by automating data preparation, model training, and evaluation pipelines, replacing manual processes.
- Developed **LLM-based solutions** using RAG and prompt engineering techniques for enterprise use cases, enhancing internal knowledge retrieval and document summarization capabilities for business stakeholders.
- Engineered **automated ML pipelines** on AWS, integrating data wrangling, experimentation, model development, and production deployment phases to support iterative model improvement.
- Integrated **ML systems** with enterprise data platform infrastructure, ensuring seamless data flow for training and inference across distributed data systems managing petabytes of data.
- Collaborated with cross-functional teams and business stakeholders to **scope and ship AI/ML systems** that solved real business problems, translating requirements into technical specifications and actionable development plans.
- Wrote **production-grade Python code** for ML model development and pipeline construction, focusing on maintainability, testability, and efficient debugging of complex data flows.
- Containerized ML services using **Docker** and deployed them to a managed Kubernetes environment on **AWS EKS**, establishing CI/CD pipelines for reliable and automated deployments.

SKILLS

AI/ML – Generative AI, LLMs, RAG, prompt engineering, machine learning, data science, anomaly detection, forecasting, model deployment, model monitoring, ML pipelines, MLOps

Programming – Python, Java, SQL

Data Platforms – Enterprise data warehouse, distributed data systems, large-scale data processing, data wrangling, data prep, feature engineering

Cloud & DevOps – AWS, Docker, Kubernetes, AWS EKS, CI/CD, Jenkins, Git, Linux

Frameworks & Libraries – LangChain, LangGraph, scikit-learn, pandas, NumPy, Spring Boot, FastAPI

BI & Concepts – BI concepts, foundational Generative AI concepts

PROJECTS

Enterprise GenAI Knowledge Assistant

- Developed a **RAG-based system** to ingest and index enterprise documentation, enabling natural language querying for internal teams and reducing information retrieval time by an estimated 40%.
- Implemented **prompt engineering strategies** within a LangChain framework to optimize LLM responses for accuracy and relevance, integrating with internal APIs for real-time data enrichment.
- Packaged the GenAI assistant as a **FastAPI service** deployed on AWS Lambda, handling approximately 150 requests per minute during peak usage periods.

ML Pipeline for Predictive Analytics

- Engineered a **Python-based ML pipeline** to automate data extraction, feature engineering, and model training for predictive analytics use cases, processing ~500 GB of data weekly.
- Trained a **scikit-learn classification model** on a dataset of 1.2 million records to forecast operational demand, achieving 0.92 AUC on a held-out test set.
- Integrated the trained model into a **CI/CD workflow** using Jenkins, ensuring automated retraining and deployment upon significant data drift detection.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)